How the OOD score gates the conditional counterfactual

Abstract

The attribution score and the OOD score answer different questions. Random-Fourier MMD^2, and the squared shift read off that kernel, say whether today's purchase law moved. The OOD score says whether the counterfactual conditioner was inside the training support of the diffusion model. A cell that fails the score does not get to speak as a headline effect. This note writes that gate down as three numbers, one action, and an eight-row evaluation. It is not a sixth frozen layer, and it does not retouch the five-layer calls from the landing case.

1 What the score is allowed to do

The conditional DDPM estimates $p(Y \mid x)$ on the x it was trained on. The counterfactual replaces the spiked IP block and rebuilds the lag vector. That new x^{cf} can leave the training marginal even when the intervention itself is the one the query asked for. Scoring Y there measures the model's extrapolation, not the conditional law the query named.

So the OOD score sits in front of the attribution, not inside it.

Object	Question	Allowed to change
Δ^{cf}, δ^2	Did the purchase law move, and by how much?	The reported effect
Support score s_∞	Is this conditioner inside the training box?	Whether the cell enters the headline
Localizer FSDS_j	Which coordinate pushed it out?	Clamp that coordinate, or drop the cell

The OOD score is not an effect size. A large FSDS_j does not mean cohort j drove purchases. It means coordinate j is why this question left the support.

2 Three numbers

The conditioner is the length-21 vector used by the denoiser: 14 IP lags, 3 covariates, overlap, and the 3 distractor blocks at lag 10. The reference cloud is the training design matrix, with coordinate mean μ and coordinate scale σ.

Support.

$$s_\infty(x) = \max_{j \le 21} \left| \frac{x_j - \mu_j}{\sigma_j} \right|.$$

The cell is in support when $s_\infty(x) \le 4$. The threshold is a coordinate box, not a joint density. A vector that is typical on every margin and unusual as a combination can pass. That is a stated limit of this score, not a hidden one.

Severity. On the standardized cloud, let $d_5(x)$ be the Euclidean distance from x to its 5th nearest training neighbor. This is the number a clamp has to reduce. It is not compared with 4. The box score and the neighbor score are two different geometries.

Driver.

$$\mathrm{FSDS}_j(x) = d_5(x) - d_5(x^{(j)}),$$

where $x^{(j)}$ replaces coordinate j by the cloud mean and leaves the rest alone. The largest coordinate is the one the fallback is allowed to touch. On a one-coordinate spike the identity $d_5(x) - d_5(x^{(j_\star)}) = \mathrm{FSDS}_{j_\star}$ is exact, and the run uses that identity as a check.

3 The chain, in order

The score empowers the attribution only by walking this list. Skipping a step is how an extrapolation gets quoted as an effect.

1. Build x^{cf} by putting the spike day back on the reference path. Later innovations, covariates, and distractors stay at their observed values.
2. Score $s_\infty(x^{\mathrm{cf}})$ against the training marginal. Do this before any MMD.
3. If $s_\infty \le 4$, the cell may enter Δ^{cf} and δ^2. The OOD score has nothing more to say.
4. If $s_\infty > 4$, compute FSDS_j on the standardized cloud.
5. One coordinate carries the distance: clamp that coordinate to the training mean, rescore, and label the cell clamped. The clamped value is a fallback, not the headline.
6. Several coordinates share the distance: drop the cell. Do not clamp a handful of them and keep the MMD.
7. The headline window is eligible only if its own conditioner passed step 3. A high support rate on other cells does not vouch for it.
8. δ^2 stays the size that gets quoted. s_∞ and FSDS_j never replace it.

4 What this run actually did

The validation grid is 10 sites, 16 horizons, 12 windows: 1920 conditioners. 1918 passed $s_\infty \le 4$, which is a rate of 0.999. Site 7 at lag 10 passed 11 of 12 windows. Site 7 is the isolated copy, and its raw score is already 0.019, so the one missed window does not move the driver call. The dose-1 headline window is inside the box. Its $\Delta^{\mathrm{cf}} = 0.674$ and its $\delta^2 = 1.192$ were not produced by an out-of-support conditioner.

The dose-response construction is the one place the code drops out-of-support rows before a concordance is computed. The site-mean Δ^{cf} still averages every window, including the one site-7 miss. On this panel that is one cell. The declaration is stricter than that average: step 7 above says the headline cell must pass on its own, and a site mean that will be read as an effect should use the same mask. That mask is not yet applied to the site table.

The localizer was not run on those 1920 real conditioners. It was run on a planted spike: lag-10 distractor b_2, coordinate 19, set to 4.5 training standard deviations, with every other standardized

coordinate at 0. FSDS named b_2 at lag 10, score 1.42, and the next two coordinates scored 0. Neighbor distance fell from 4.05 to 2.63 after clamping that coordinate. The difference is 1.42, which matches the score. The residual 2.63 is the 5th-neighbor distance of the cloud mean; clamping does not make a point identical to a training row. The real attribution path never entered this branch.

5 Evaluation list

Each row is a check on the gate. "This run" is the linear ten-site validation, not a new experiment. A fail here does not rewrite L1–L5. L5 still fails on the raw ratio, and L3 still fails on Wasserstein.

	Check	Pass condition	This run	Call
1	Coverage	Share of real x^{cf} with $s_\infty \le 4$ is reported, and the headline cell is among them	1918/1920; headline in support	pass
2	Gate binds	Out-of-support cells are counted, and the dose construction ignores them	2 cells out; dose uses the mask	pass
3	Site means	Any Δ that is read as an effect uses the same mask	Site table still averages the site-7 miss	gap
4	Planted driver	Argmax of FSDS is the planted coordinate	b_2 at lag 10, score 1.42	pass
5	Clamp identity	d_5 falls by $\mathrm{FSDS}_{j_\star}$ and by nothing else	$4.05 - 2.63 = 1.42$	pass
6	Real-path localizer	FSDS is computed on real out-of-support cells, not only on a plant	Not run; the two real misses were not localized	gap
7	Separation	δ^2 is the quoted size; s_∞ is not	δ^2 grid $0.808, 0.798, 0.787$; oracle mean of squares 0.754	pass
8	Joint hole	State that s_∞ is a box. A joint OOD point with small margins is a known miss	Stated. Not probed on this panel	open

Rows 3 and 6 are the two places the written chain is ahead of the code. Neither one changes the driver on this panel: the missed cell is a negative control, and the headline did not need a clamp. They are still listed, because a later panel with a real excursion would otherwise quote an extrapolated MMD as a site effect and would have no coordinate to clamp.

6 What stays put

The five-layer card is unchanged. Primary driver: site 0, lag 10, IP block, weekly scale. Headline $\Delta^{\mathrm{cf}} = 0.674$ with interval $[0.442, 0.906]$. Squared shift across the bandwidth grid: coefficient of variation 0.01. Raw ratio across that grid: 8.9 to 46.4, and that clause still fails. Feature-only Wasserstein floor: 0.166, model 0.245, and the 0.10 line still sits under the floor.